

Cybersecurity

Introduction to Networking

Lesson 8.1



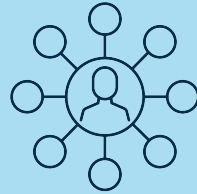
Learning Objectives

By the end of today's lesson, you will be able to:

- 1 Identify clients, servers, requests, and responses in network communications.
- 2 Identify three benefits of AI/ML for networking.
- 3 Design a conceptual network made of various networking and network security devices.
- 4 Convert binary numeric representations to readable IP addresses and determine which servers the IP addresses belong to.
- 5 Modify hosts files to circumvent DNS and redirect the access of a website.



Welcome



Intro to Networks and Network Security

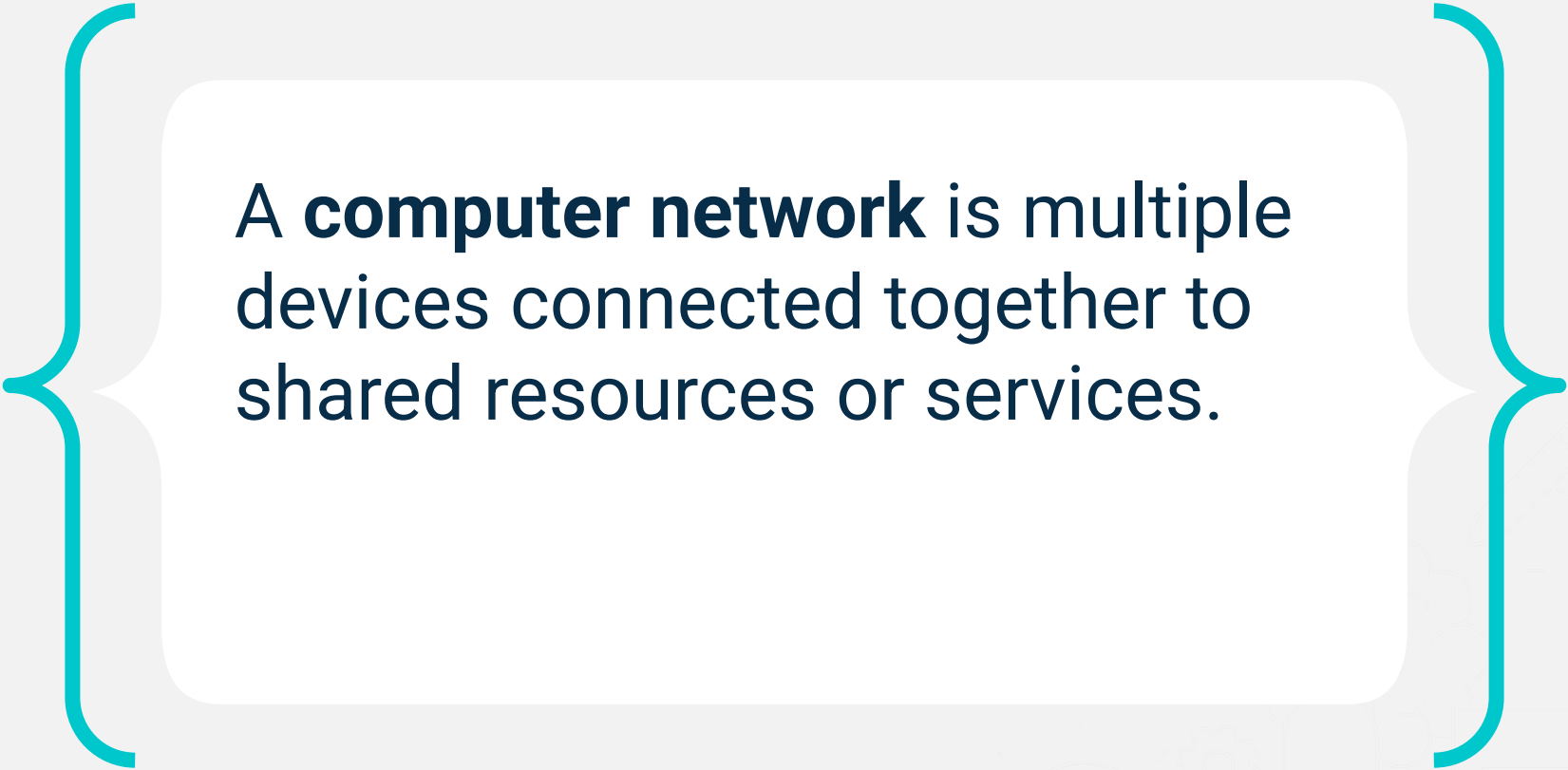
Today, we will be
switching gears
to focus on one of the
most important topics
in cybersecurity:
**computer
networking.**

```
..._mod = modifier_ob.  
... mirror object to mirror  
... mirror_mod.mirror_object  
... operation == "MIRROR_X":  
... mirror_mod.use_x = True  
... mirror_mod.use_y = False  
... mirror_mod.use_z = False  
... operation == "MIRROR_Y":  
... mirror_mod.use_x = False  
... mirror_mod.use_y = True  
... mirror_mod.use_z = False  
... operation == "MIRROR_Z":  
... mirror_mod.use_x = False  
... mirror_mod.use_y = False  
... mirror_mod.use_z = True
```

```
... selection at the end -add  
... _ob.select= 1  
... _ob.select=1  
... context.scene.objects.active  
... ("Selected" + str(modifier_ob.  
... mirror_ob.select = 0  
... bpy.context.selected_object  
... data.objects[one.name].select  
... print("please select exactly one")
```

```
... OPERATOR CLASSES ----  
... types.Operator):  
... X mirror to the selected  
... object.mirror_mirror_x"  
... mirror X"
```

```
... context):  
... context.active_object is not
```



A **computer network** is multiple devices connected together to shared resources or services.



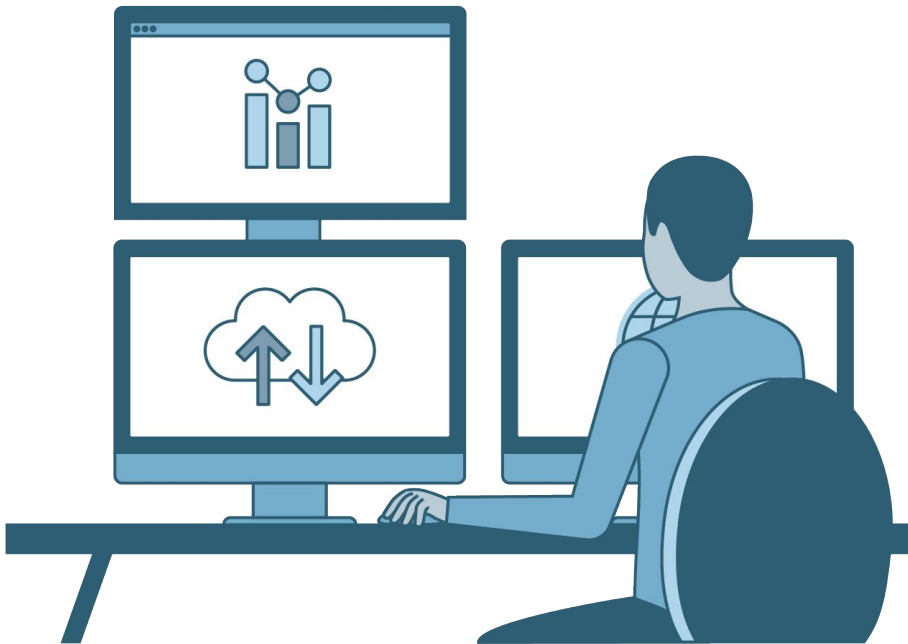
Computer Networks





Networks and Network Security

Computer networking is essential for:



Security Operations Center (SOC) Staff

- Diagnose and troubleshoot network-related security issues and attacks.



Networks and Network Security

Computer networking is essential for:



Network Security Engineers

- May work on the design of a company's network architecture to protect their organization from security risks.



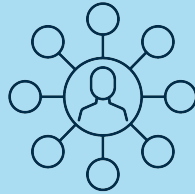
Networks and Network Security

Computer networking is essential for:

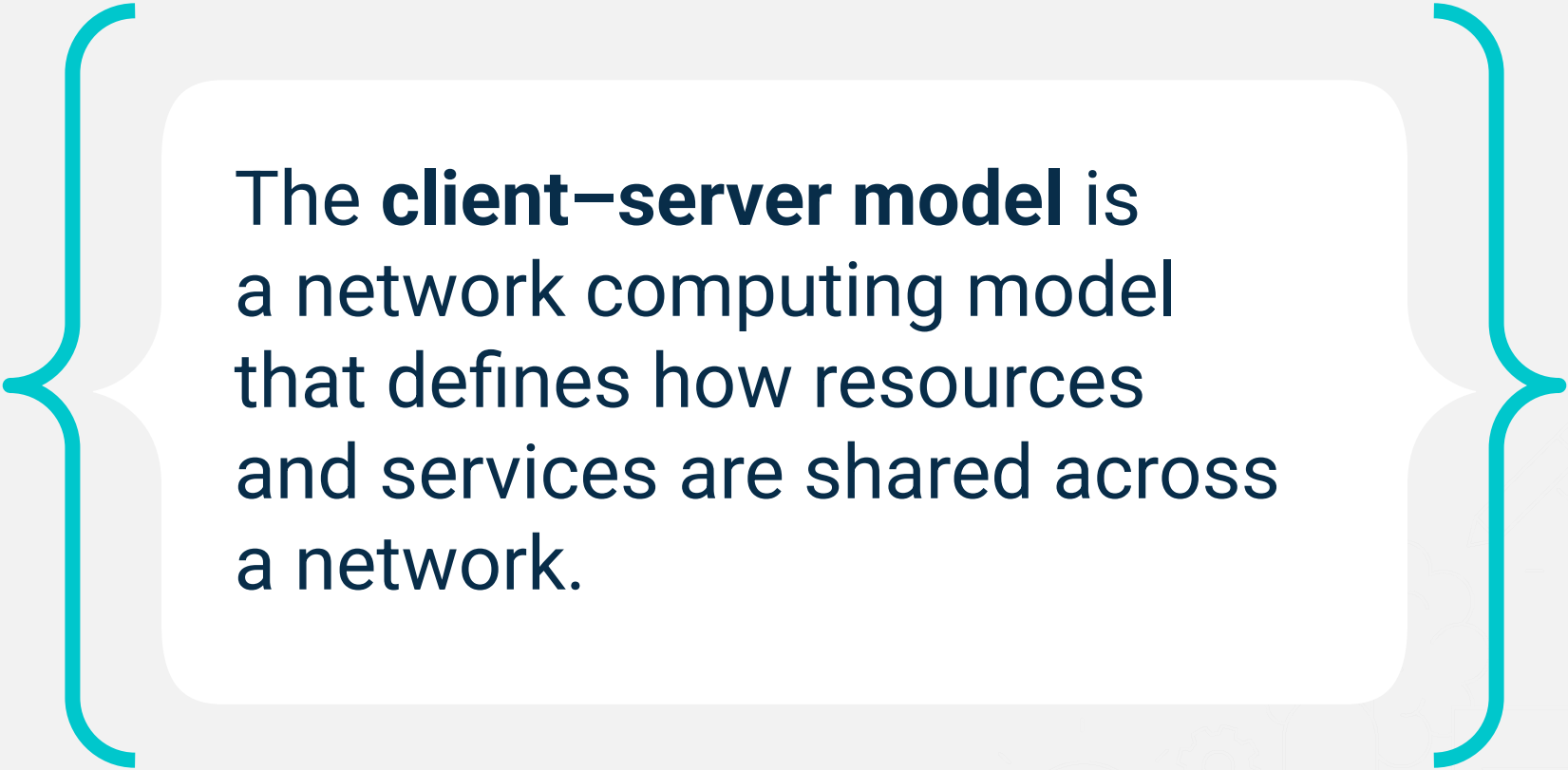


Penetration Testers

- Test for vulnerabilities in a company's network.



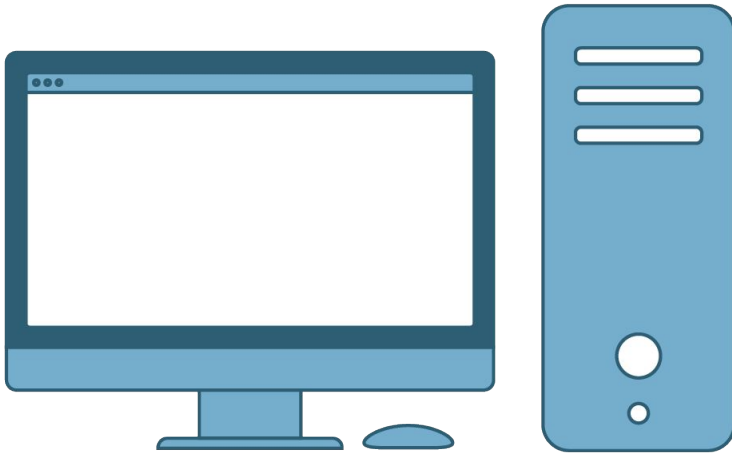
The **Client–Server** Model



The **client-server model** is a network computing model that defines how resources and services are shared across a network.

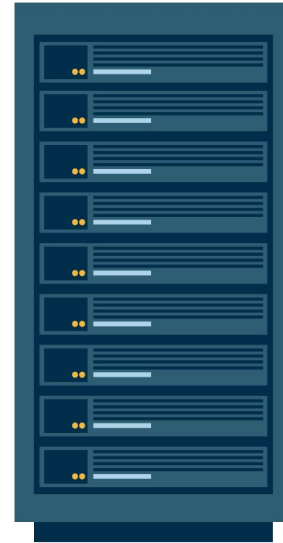
Client

The client requests a resource or a service.



Server

The server responds with the resource or service.



Client and Server Example

Let's take the example of a webpage.

You want to view the vacation photos your friend posted on Facebook. When you view them ...

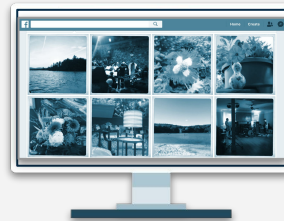


Request

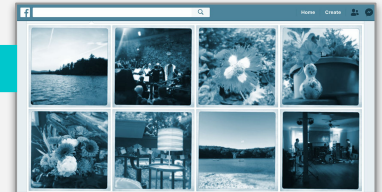
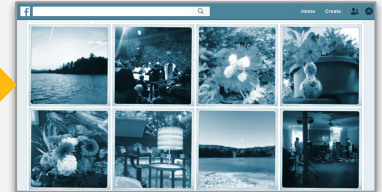
... the browser is making a request: *"Facebook, can you please get me my friend's vacation photos?"*



Return



Facebook's web server provides a response: *"Yes, here are your friend's vacation photos."*





This **two-way conversation** between the client and server is known as the **request and response method** of device communication.

Client-Server Model

01

Request:

Process in which the client sends a message to a server asking for a resource or to run a service.

02

Response:

Message sent back to the client after the server receives and processes the request.

03

Response Message:

- An acknowledgement of the request
- A requested resource
- An error message

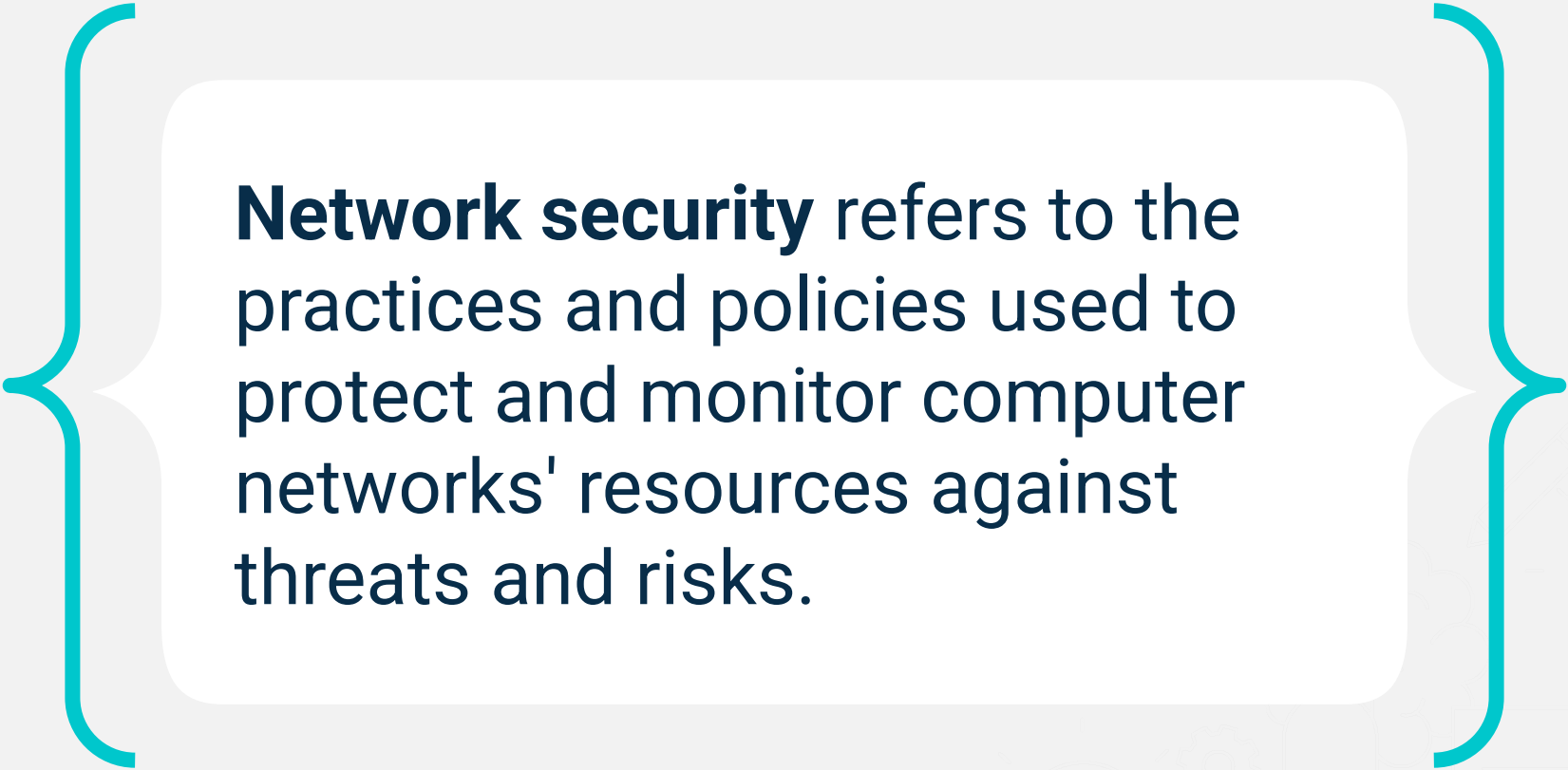
One Server, Many Clients

The client-server model is not a “one client, one server” relationship. Typically, servers receive resource requests from many clients.





Introduction to **Network Security**



Network security refers to the practices and policies used to protect and monitor computer networks' resources against threats and risks.

Network Security

- Unauthorized access to networks
- Denial of service (DoS) attacks
- Eavesdropping
- Data modification



As security professionals, you will often be asked to not only monitor and identify potential network security threats and risks, but also to determine the best way to mitigate them.





Questions?





Activity

Network security activity

In this activity, you will play the role of a security analyst at Acme Corp.

- Several suspicious activities have occurred on Acme's products and systems.
- For each suspicious activity, you must list out the client and servers, and then describe the request and response in simple terms.

Suggested Time
10 minutes

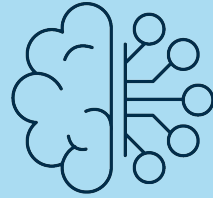


Time's Up!
Let's review



Questions?





Networking and **Artificial Intelligence**

Due to the growth of emerging technologies, there are many more complexities involved in protecting networking environments.

Organizations try to overcome these challenges by leveraging **Artificial Intelligence (AI)** and **Machine Learning (ML)**.



Networking and Artificial Intelligence

AI/ML can assist with networking challenges.

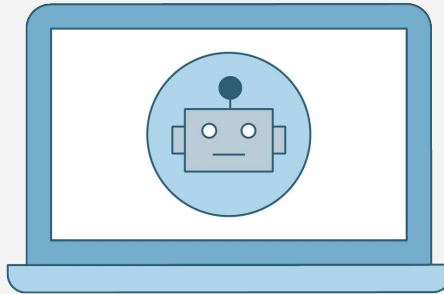
01

**Troubleshooting
Network Issues**



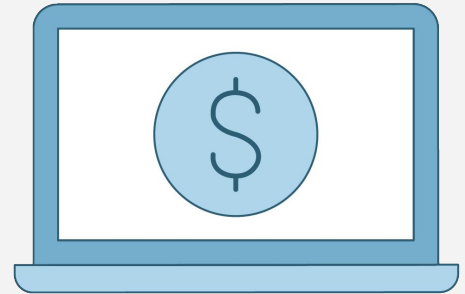
02

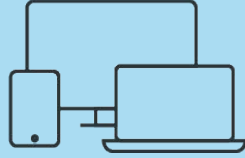
**Self-Resolving
Networking Issues**



03

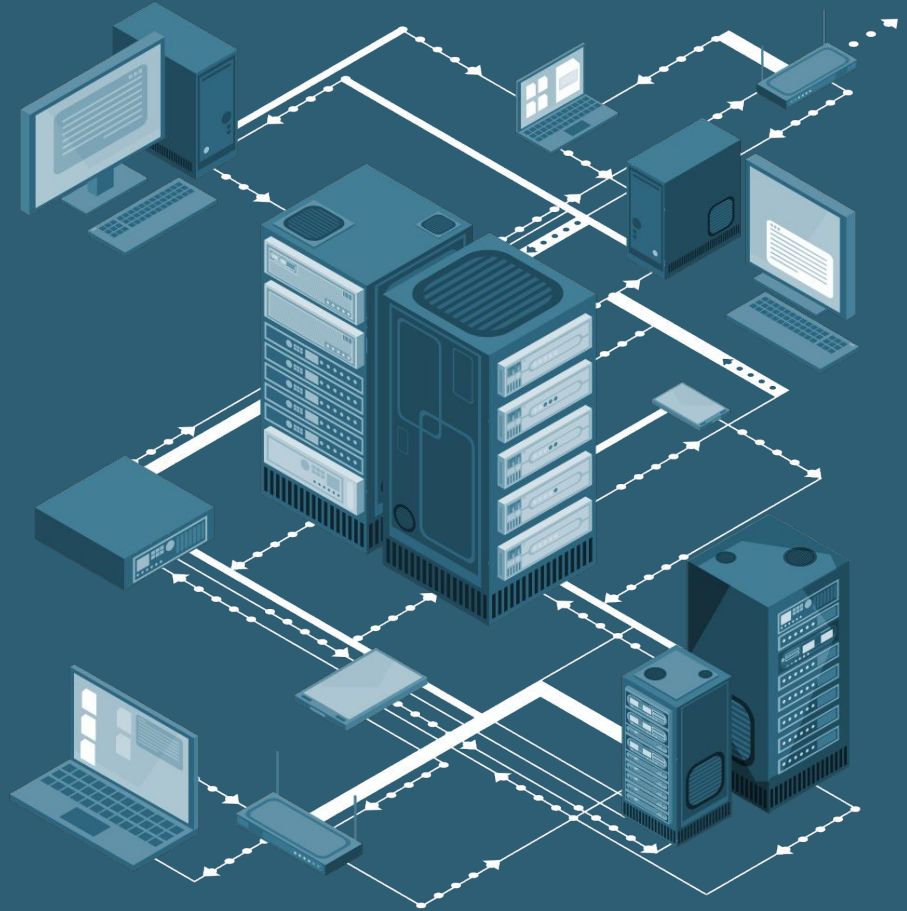
**Organizational
Network Cost Savings**





Network Devices

The **nodes** that connect a network are actually many different devices, each with complex responsibilities.





We will now cover
the **primary network
devices** found on
LANs and WANs.





Primary Network Devices

Router



A **router** is a networking device that forwards (routes) resources to other networks.

Routers connect:

- Two LANS
- Two WANS
- A LAN to a WAN

Routers can use **AI/ML** to determine optimal routes.



Primary Network Devices

Switch

A **switch** is a networking device that forwards resources within a network.

Switches:

- Connect devices in a LAN
- Used in large businesses
- Feed into routers
- Intelligent devices

Switches can use **AI/ML** to become energy efficient by disabling any unused connections.





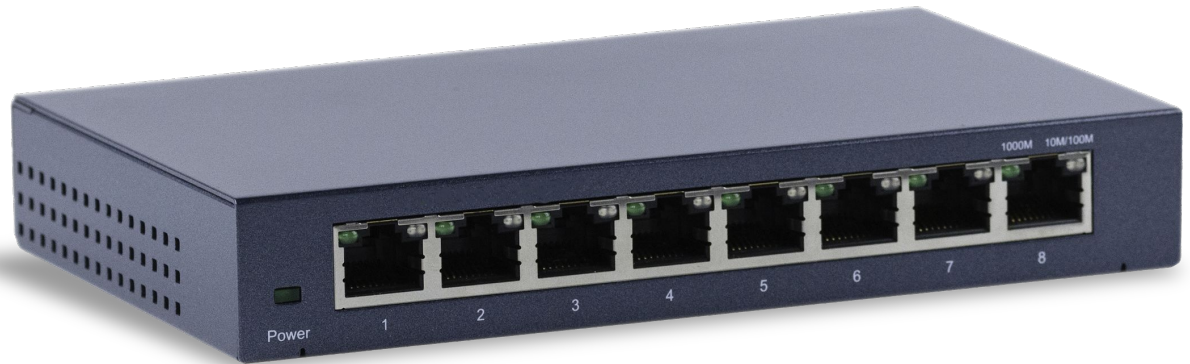
Primary Network Devices

Hub

A **hub** serves the exact same purpose as a switch, except it is not an intelligent device.

Hubs:

- Can't be programmed
- Less secure



Hubs are outdated and no longer commonly used.



Primary Network Devices

Bridge



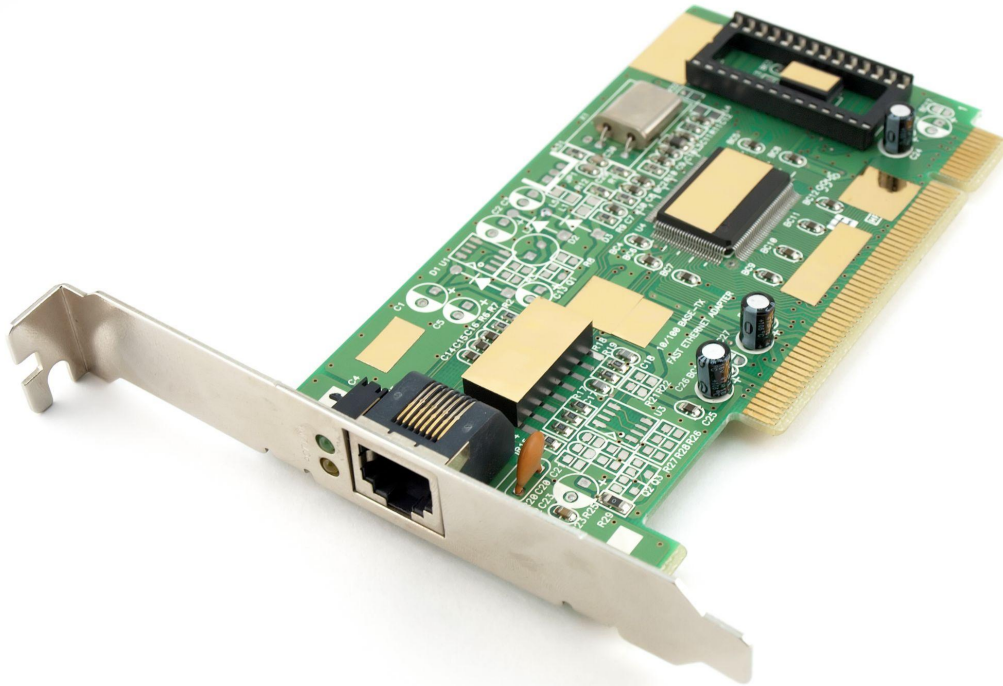
A **bridge** is basically a switch that only has two connections—one in and one out.

- **Bridges** are often used to tie two LANs together.



Primary Network Devices

Network interface controller (NIC)



A **NIC** is a type of computer hardware that connects a computer to a computer network.

NICs:

- Usually a circuit board or chip
- Need an NIC to receive or send
- Both wired and wireless



Primary Network Devices

Modem

A **modem** converts resource data into a format that the next type of connection can understand.

Modem:

- Short for modulator-demodulator
- Translates between your computer and the ISP



Network Devices: Wireless Access Point (WAP)

WAPs give wireless devices the ability to connect to a wired network.





Primary Network Devices

All-in-one device

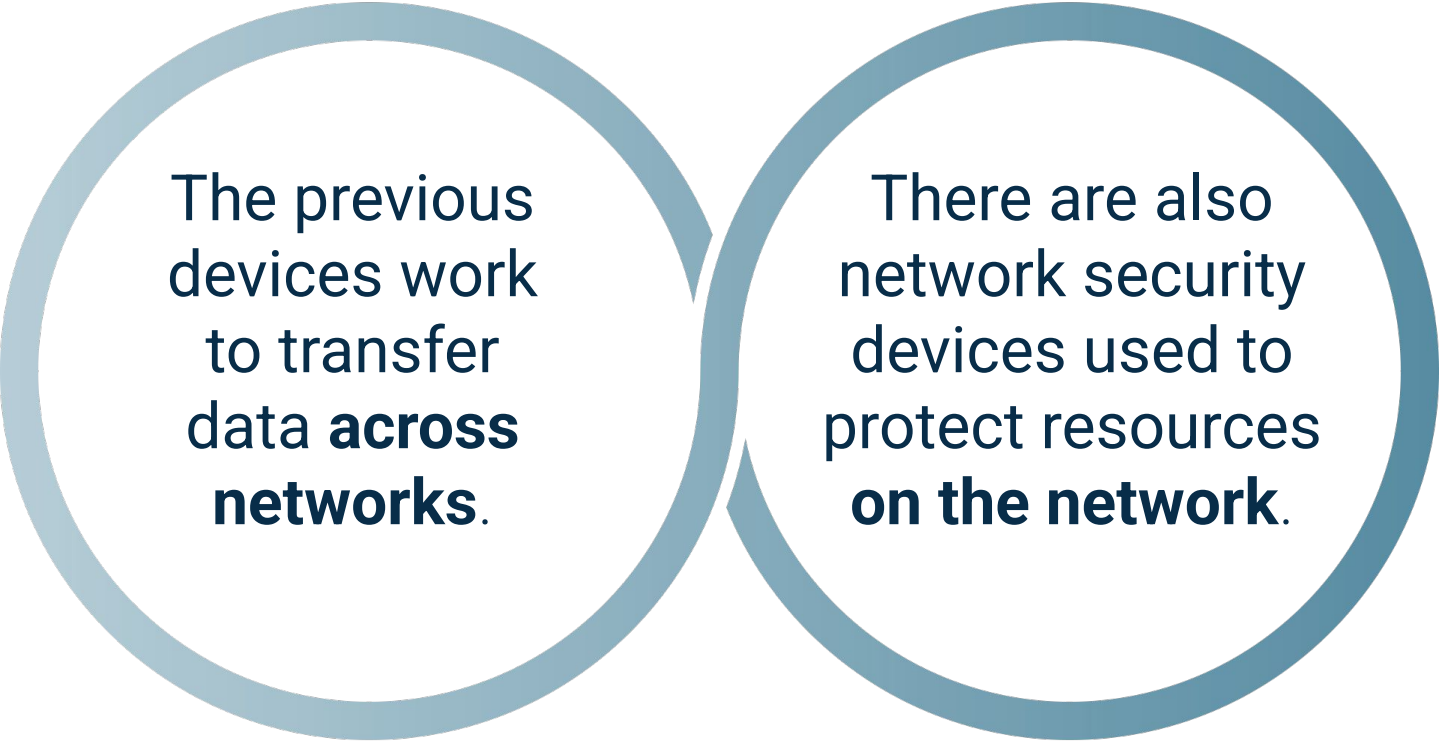


- Easy to use
- Less to set up



- Single point of failure
- Difficult to troubleshoot





The previous
devices work
to transfer
data **across**
networks.

There are also
network security
devices used to
protect resources
on the network.



Primary Network Devices

Firewall

A **firewall** is an intelligent network security device that monitors incoming and outgoing traffic based on security rules.

Firewall:

- Typically placed at entry point

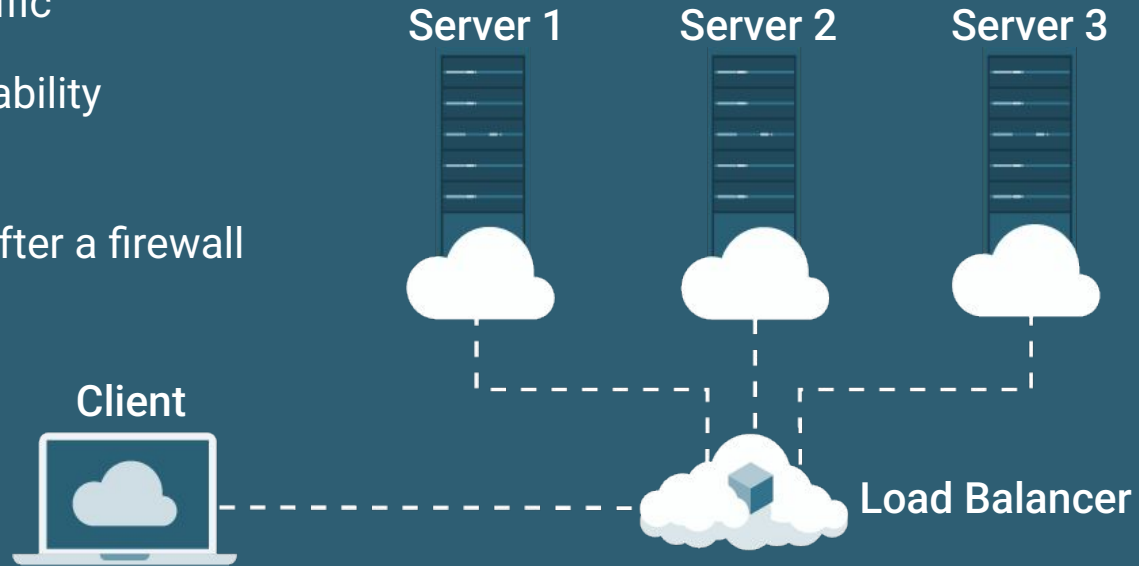


There are many types of firewalls and specific firewall functionalities, many of which utilize **AI/ML** and will be covered in more detail in future lessons.

Network Security Devices: Load Balancer

A **load balancer** is an intelligent network security device that distributes incoming network traffic across multiple servers.

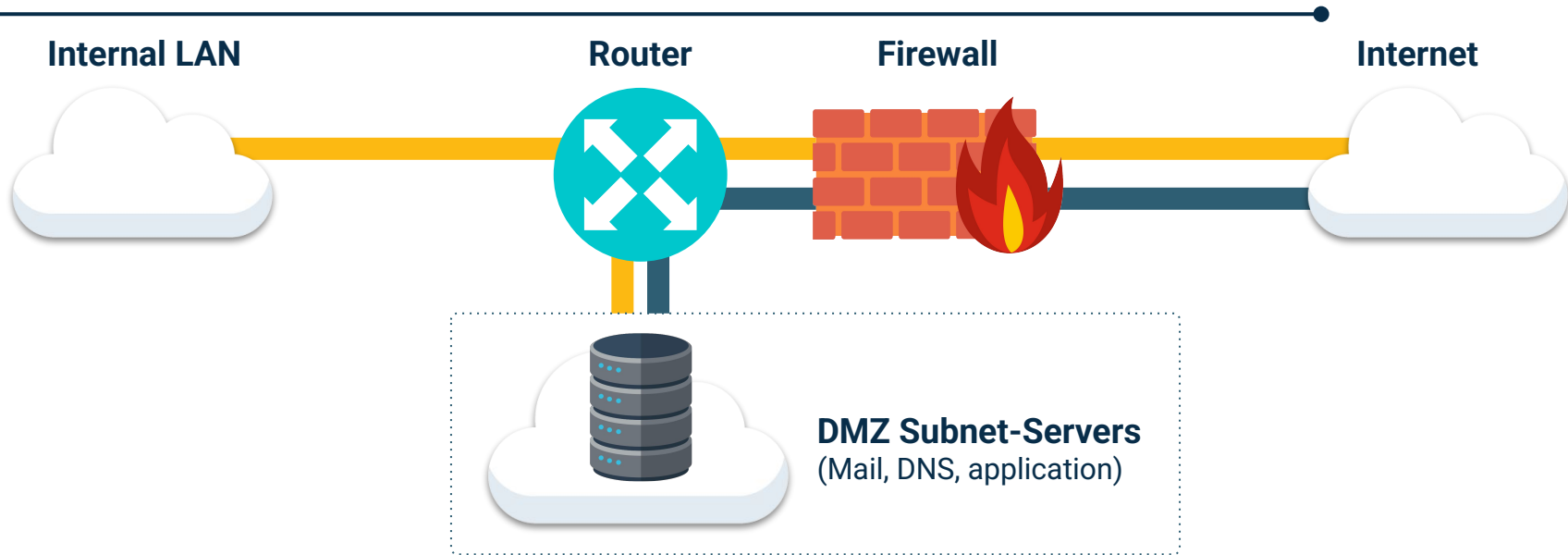
- Ensures no single server has to handle too much traffic
- Helps protect the availability of resources
- Typically placed right after a firewall





Primary Network Devices

DMZ



- A **DMZ** is a smaller subnetwork, usually within a LAN
- Additional layer of security
- Typically has its own network security devices

A common task for network and security professionals is to **visually design a setup** before purchasing, installing, and configuring a network with these devices.



Network Visualization

Visually designing your network can assist with the following:

01

Making networks more efficient, since proximity of certain devices can reduce latency.

02

Avoiding the creation of a “single point of failure.”

03

Ensuring private resources are protected from unauthorized sources.

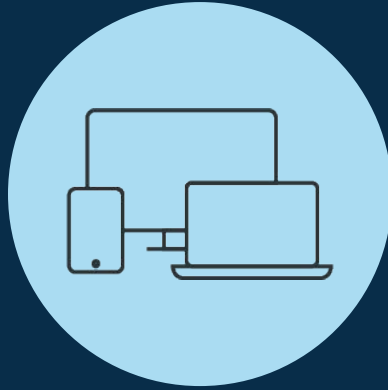


Demonstration Scenario



We will use the free web tool **draw.io** to design a basic network incorporating the following devices:

- Two computers
- One router
- One switch
- One firewall
- One representation of the internet



Instructor Demonstration

draw.io



Activity

Network security activity

In this activity, you will continue to play the role of a security analyst at Acme Corp.

- Acme Corp just opened a new office in Shanghai, China.
- Your task is to design the network layout for the Shanghai office using **draw.io**.

Suggested Time
25 minutes



Time's Up!
Let's review



Questions?





Break
15 mins

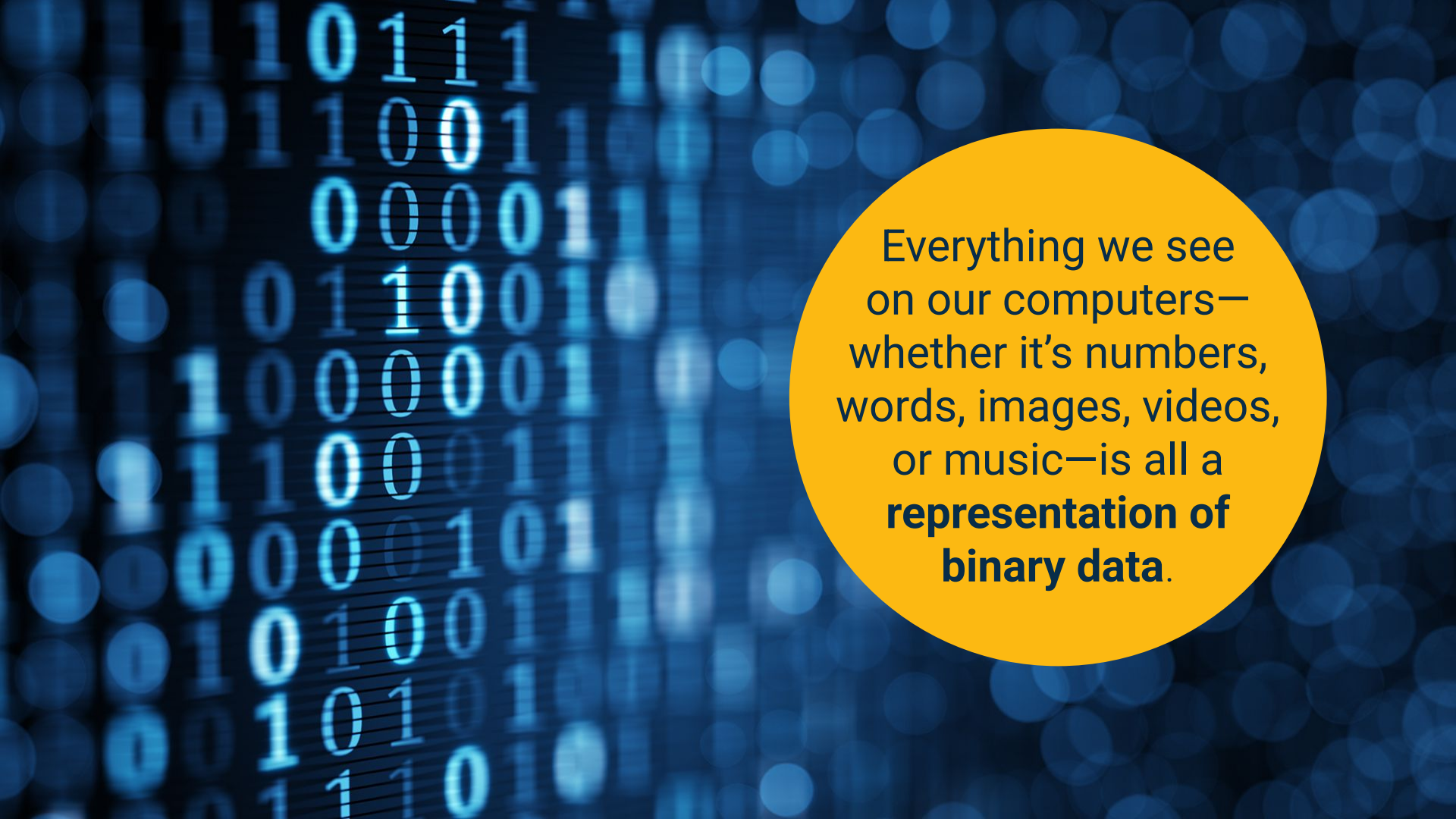


What's my (**Network**) Address?



Computers and
networks **don't**
communicate the
same way people do.

We use
a language
called **binary.**



Everything we see
on our computers—
whether it's numbers,
words, images, videos,
or music—is all a
**representation of
binary data.**

Binary Data

At the lowest level, computers communicate with electrical signals, which have two states: **on** and **off**.

Binary is a **two-digit** numerical system that computers use to communicate.

1 signifies an **on** signal.

0 signifies an **off** signal.

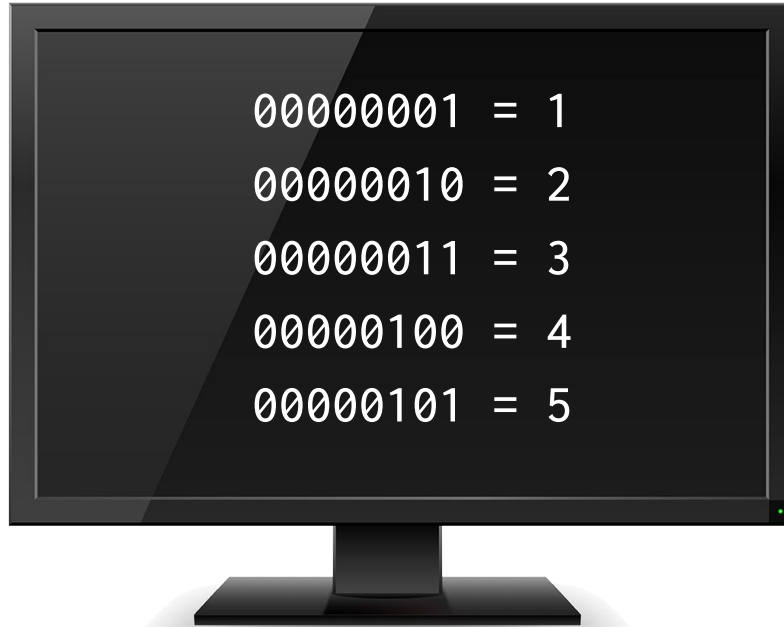
Computers transmit these electrical signals from one computer to another, and the signals get converted into **binary data**.



Binary Data

Example: If one computer wants to transmit 1 2 3 4 5 to another computer, it can't simply transmit these five numbers as we read them because computers only speak in binary.

The computer
would transmit
the binary data:



The conversion of this binary
data into a numerical
representation of

1 2 3 4 5

is called **binary to decimal**.

Receiving Computers Use Other Conversions, Such As:

Binary to ASCII

ASCII is primarily used to convert binary to readable text.
`01101000 01101001 = hi`

Binary to Hexadecimal

Hexadecimal, or hex, shortens binary to letters and numbers.
`11000111 00000110 10100110 11100110 11110110 01000110`
C7 6 A6 E6 F6 46

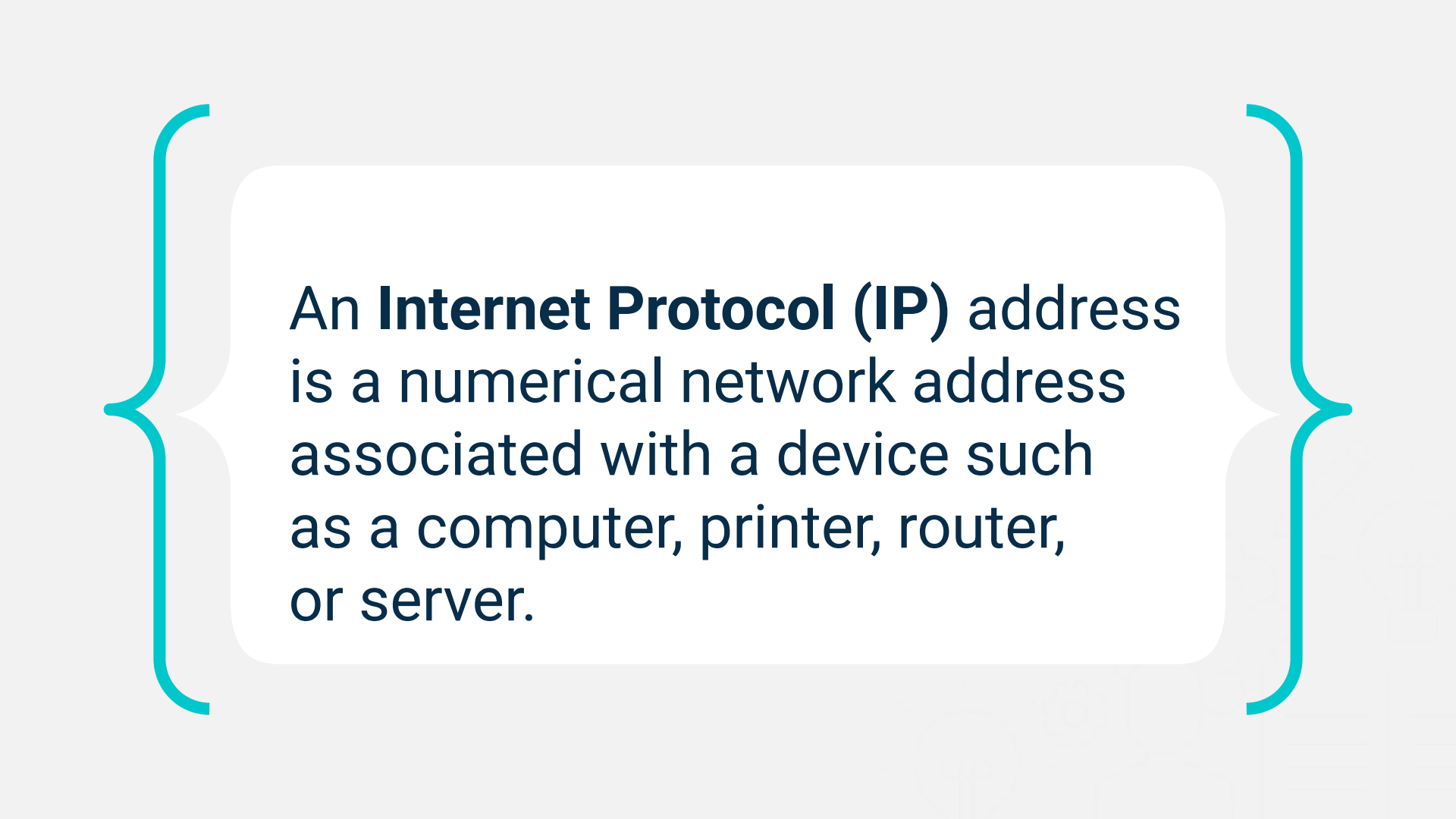
Binary to Octal

Octal is another way to shorten binary data with numbers.
`11000111 00000110 = 307      6`

Binary and Network Addresses

A network address is similar to a mailing address. Without a mailing address, we wouldn't know where to send mail.





An **Internet Protocol (IP)** address is a numerical network address associated with a device such as a computer, printer, router, or server.

Each of our computers has an IP address.
Let's check them at: **www.whatsmyip.org**



IP Addresses

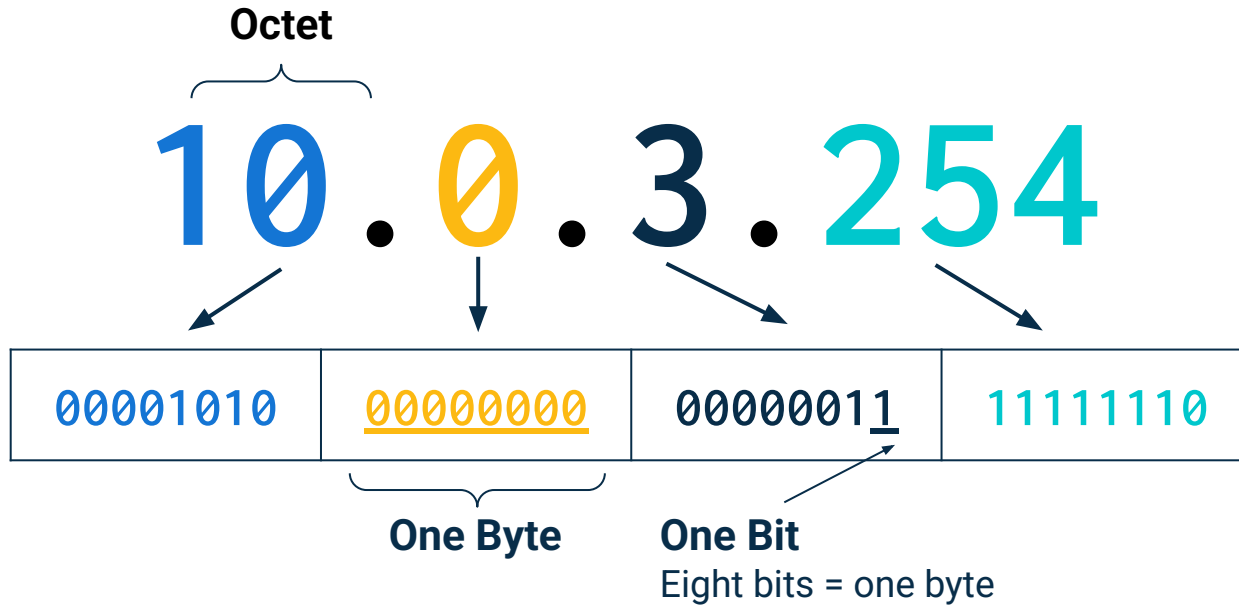
Internet Assigned Numbers Authority (IANA)

- Sub-organizations responsible for the distribution of IP addresses
- Two primary versions of IP addresses:
 - IPv4
 - IPv6

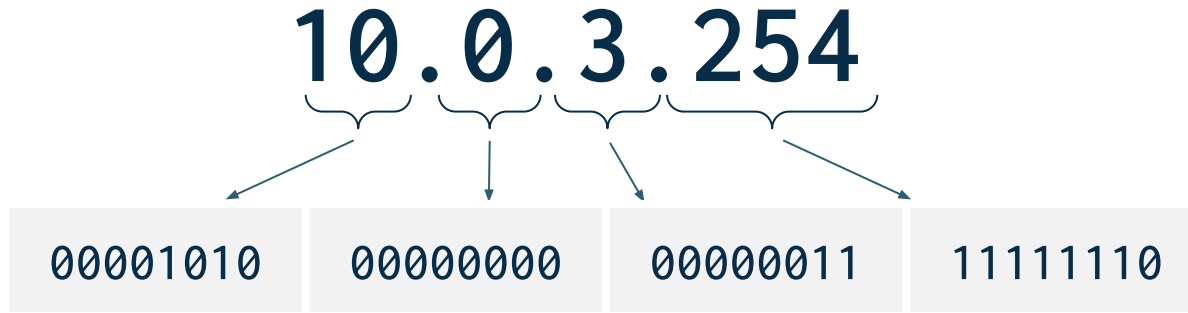


IPv4

- Four octets separated by decimals
- Conversion of **eight binary bits** or **one byte**



IP Example



What computers read:

00001010.00000000.00000011.11111110



What we read:

10.0.3.254

Each Octet can range from 0 - 255

Lowest value of eight bits

00000000

=

0

Highest value of eight bits

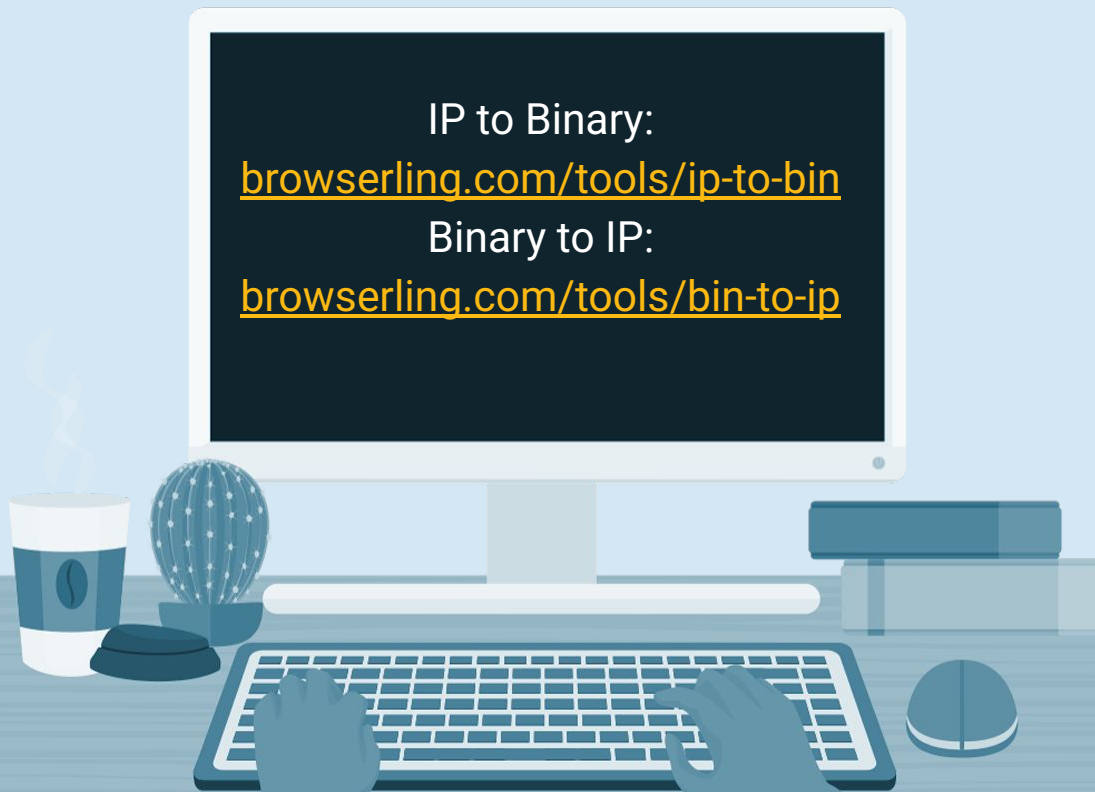
11111111

=

255

Converting IP

Converting IP addresses can be tricky. Fortunately, there is a web tool that can easily convert IP to binary and binary to IP.



IPv6

There is another version of IP addresses called IPv6 (IP version 6).

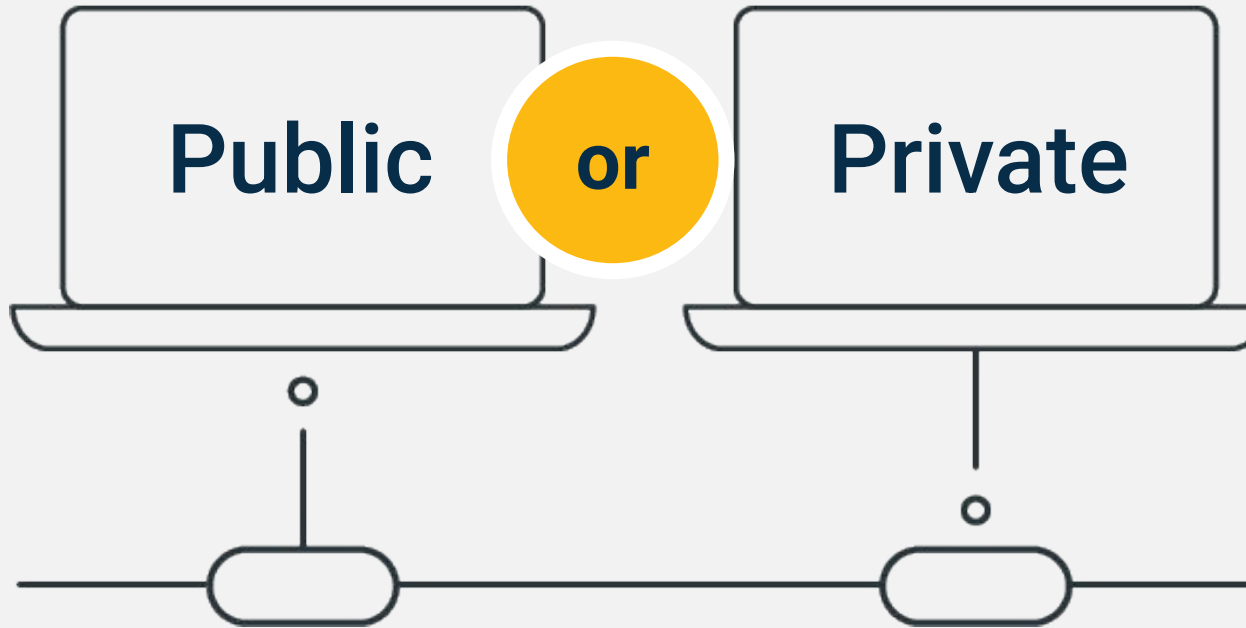
- IPv6 was created due to concern about the lack of possible addresses provided by IPv4.
- IPv6 addresses are divided into eight groups of two bytes.
- IPv6 has not yet been widely adopted. Many devices need to be updated before accepting and sending traffic with IPv6.

2001 : 0db8 : 85a3 : 0000 : 0000 : 8a2e : 0370 : 7334



- For this reason, we won't be discussing IPv6 in more detail, but it is still important to know that there are two possible IP versions.

IP addresses are categorized as:





Public IPs

Advantages:

- Accessible over the internet

Disadvantages:

- Not all devices should be exposed

Public IP addresses are typically assigned in **IP ranges** by an ISP.

- 108.0.0.1 - 108.0.0.3
 - 108.0.0.1
 - 108.0.0.2
 - 108.0.0.3

Private IPs

Advantages:

- More secure
- Reused
- No confliction

Disadvantages:

- Not directly accessible
- Assigned by LAN



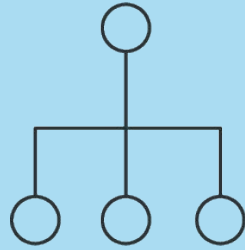
Private IPs

Three IPv4 ranges for private IP addresses.

Starting IP	Ending IP	IP Addresses Available
10.0.0.0	10.255.255.255	16,777,216
172.16.0.0	172.31.255.255	1,048,576
192.168.0.0	192.168.255.255	65,536



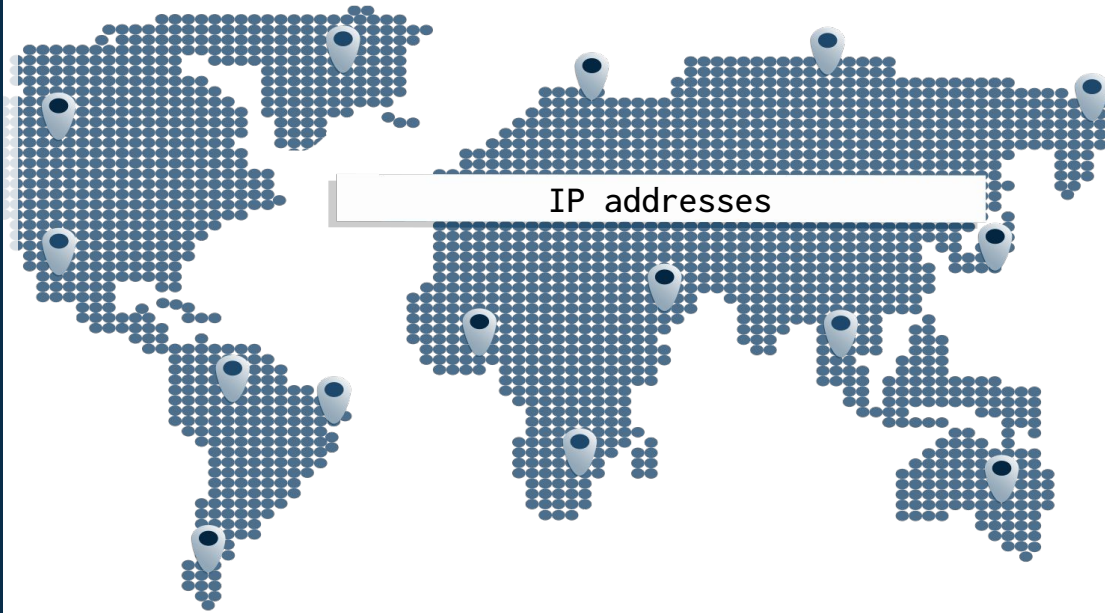
- All addresses **not** in these three ranges are considered public.



Subnetting

IP addresses are assigned manually by the user or organization that manages their local network.

But how do organizations decide what IP addresses are assigned?



Subnetting

Grouping devices together on a network for organizational and efficiency.

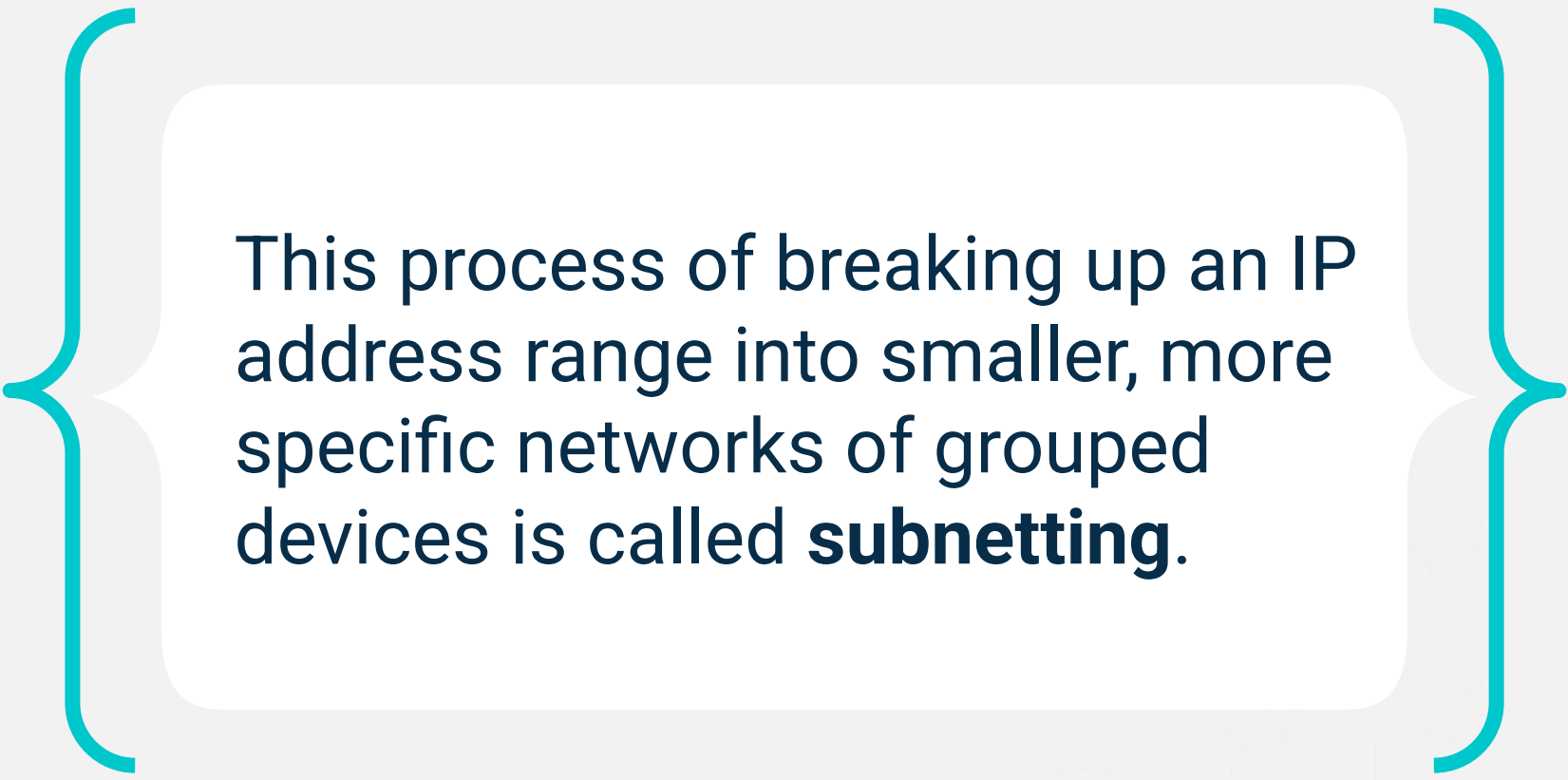
For example:

Organizations are typically provided a range of IP addresses that they will distribute across departments and devices.



An organization would group together servers designated for finance and servers designated for marketing.



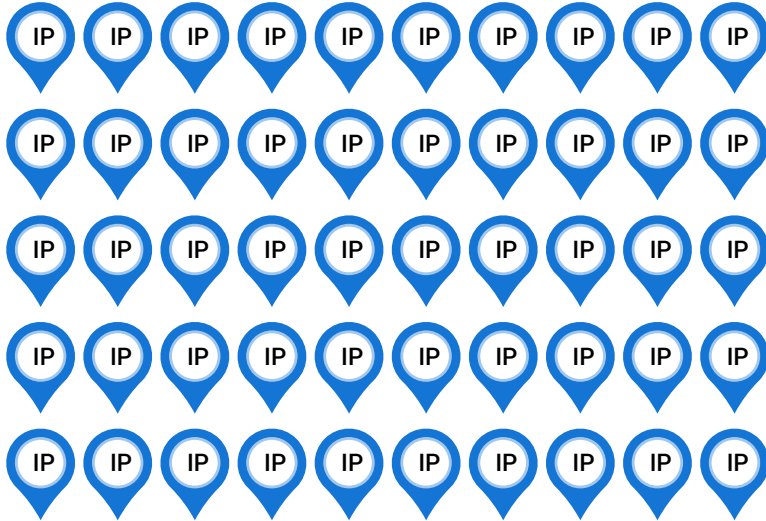


This process of breaking up an IP address range into smaller, more specific networks of grouped devices is called **subnetting**.

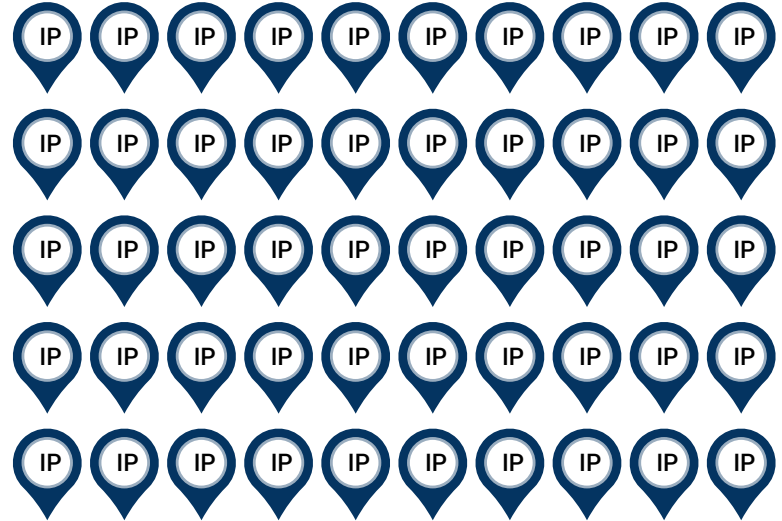
Subnetting

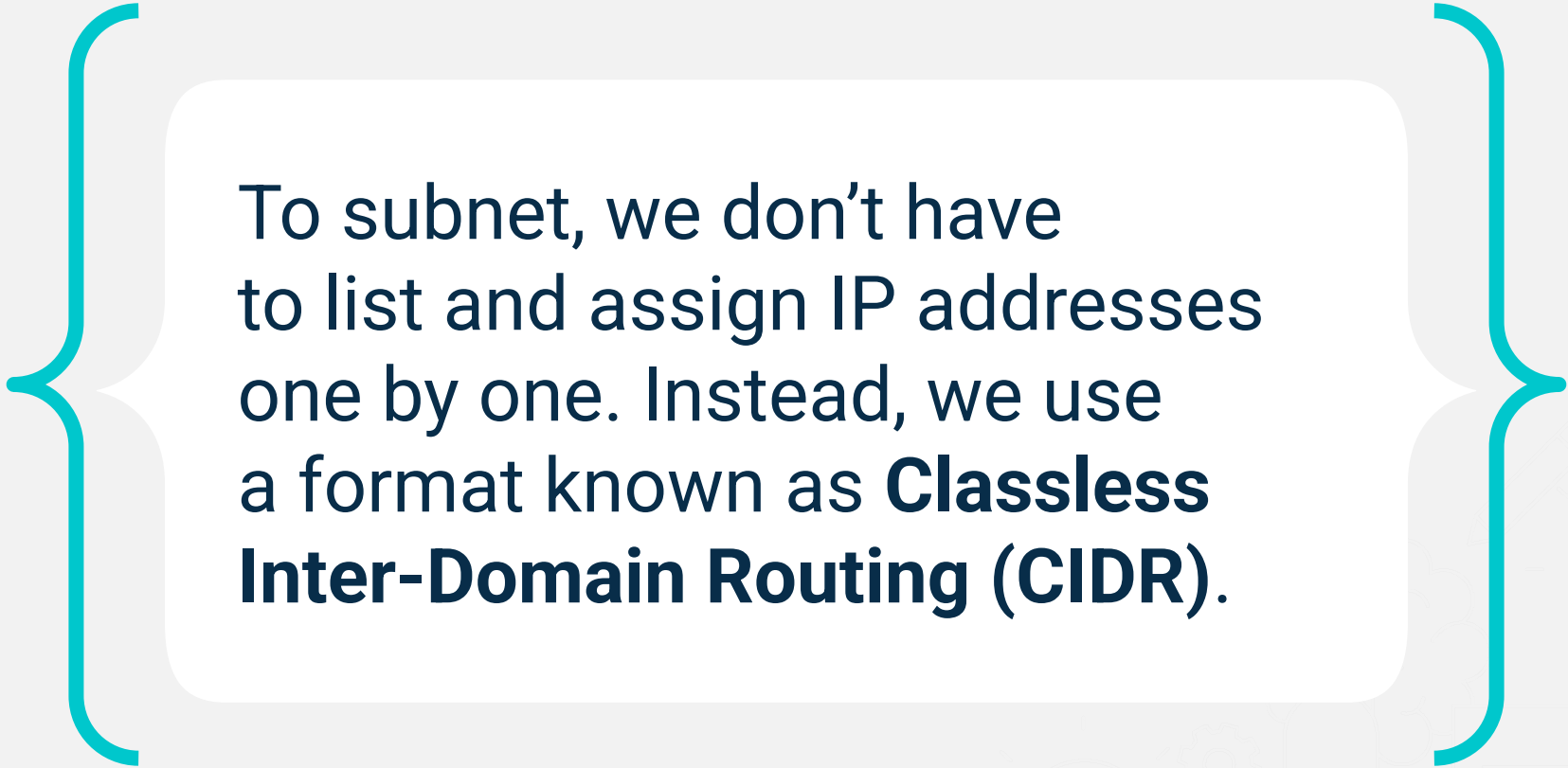
Example: If a company has 100 new IP addresses to distribute, it can assign 50 to the finance department and 50 to the marketing department by subnetting its provided IP range.

Finance



Marketing





To subnet, we don't have to list and assign IP addresses one by one. Instead, we use a format known as **Classless Inter-Domain Routing (CIDR)**.

CIDR

The CIDR format consists of two sets of numbers:
an IP address (the **prefix**) and a range of available IP addresses (the **suffix**).

192.243.3.0

Prefix:

The IP address.

The **first 3 octets** of
this prefix would be
the network ID.

/24

Suffix:

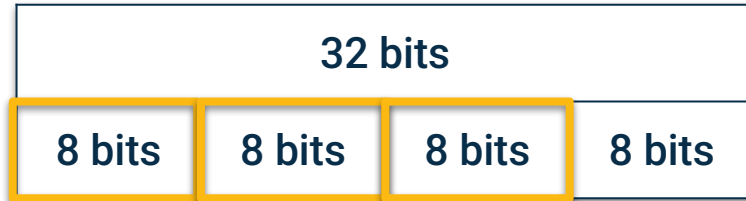
The range of IPs and the
number of IPs available.

24 means everything
after the first 24 bits
is **variable**.

CIDR

The CIDR format consists of two sets of numbers:
an IP address and a range of available IP addresses.

192.243.3.0 /24

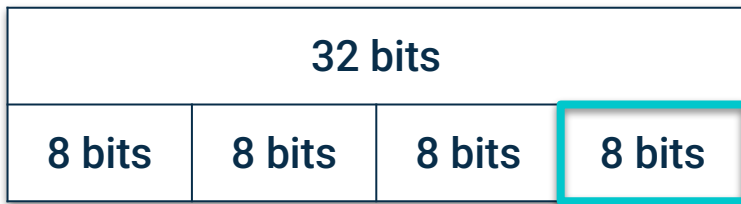


24 means the first 24 bits
(three octets) are **static**
for a given network.

CIDR

The CIDRs format consists of two sets of numbers:
an IP address and a range of available IP addresses.

192.243.3.0 /24

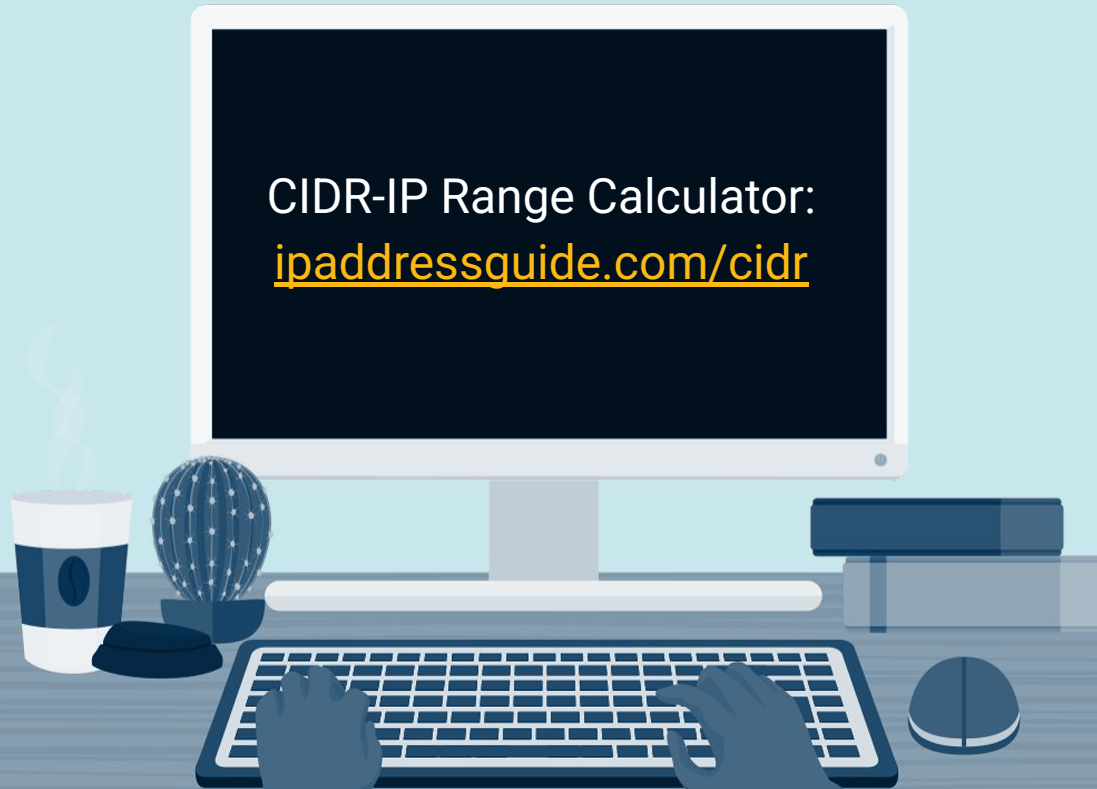


The last 8 bits (.0) is **variable**.

- The range of each octet is 0–255.
- There are 256 available IP addresses in the range 192.243.3.0/24.
- 0 and 255 are reserved. 0 is reserved for the subnet ID and 255 is reserved for the broadcast.

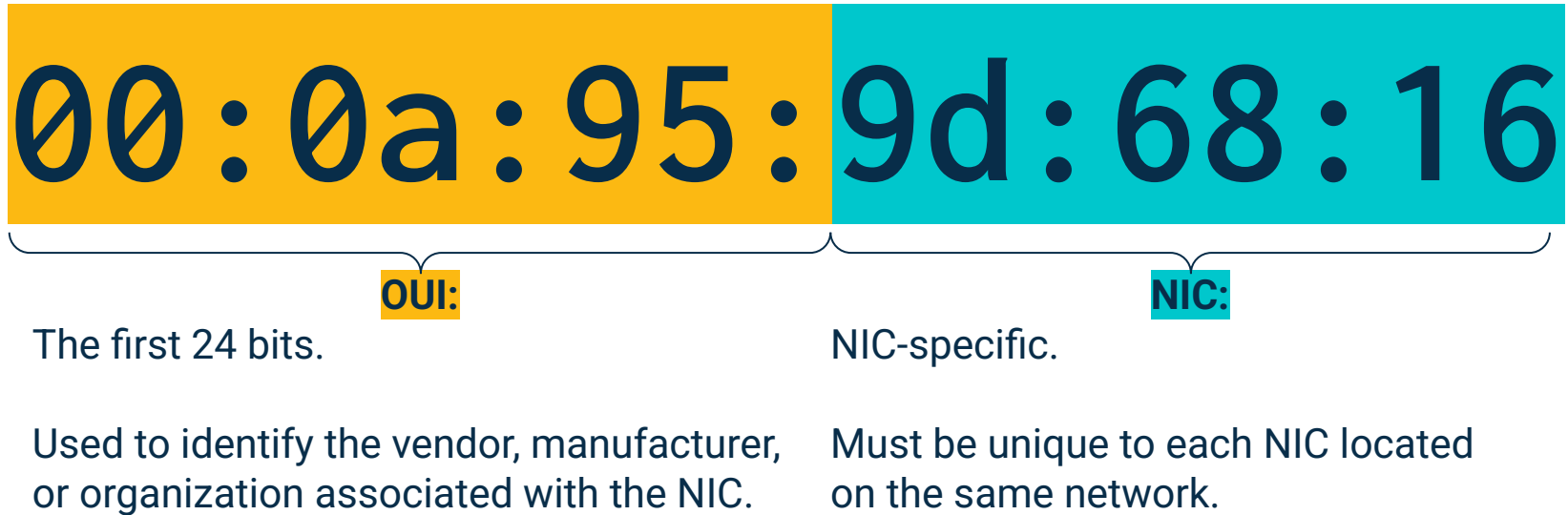
Range Calculator

We can use online tools to easily calculate an IP address range.



Media Access Control (MAC) Address

- A MAC address is a string of six sets of alphanumeric characters, separated by colons.
- MAC addresses are burned-in addresses assigned to network interface cards (NICs).





Activity

Network addressing activity

In this activity, you will continue to play the role of a security analyst at Acme Corp.

- Your task is to convert the binary traffic into an IP address, determine if it is a public or private address, and compare the IP to a list of Acme's servers to see which system the hacker is trying to access.

Suggested Time
20 minutes



Time's Up!
Let's review



Questions?





Addresses and the Internet

Addresses and the Internet

We covered how network addresses assist with directing traffic to its final destination. When you visit a website, a similar process occurs.

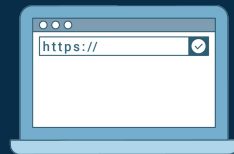


DNS

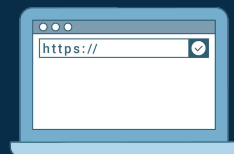
Domain Name System

It would be challenging to remember the IP 31.13.65.36 every time we wanted to visit **facebook.com**.

Thankfully, we just have to remember "facebook.com," because the **DNS translates** the domain of the website into an IP address.



31.13.65.36



facebook.com





DNS

Phone book of the internet

When you visit a webpage on your browser, your browser is looking up the associated IP address of the domain behind the scenes.

This process is called **DNS lookup**.

When a website is entered in a browser, the browser will check DNS caches.

Layers of Cache

Caches are searched in ascending order.

01

The browser's cache

02

The operating system's cache, stored in the host's file

03

The internet service provider's (ISP) cache

04

The top-level domain (TLD) cache



URLs

Uniform Resource Locators

A domain is the website we access for resources. The resources we're requesting are typically at a specific location within that domain.



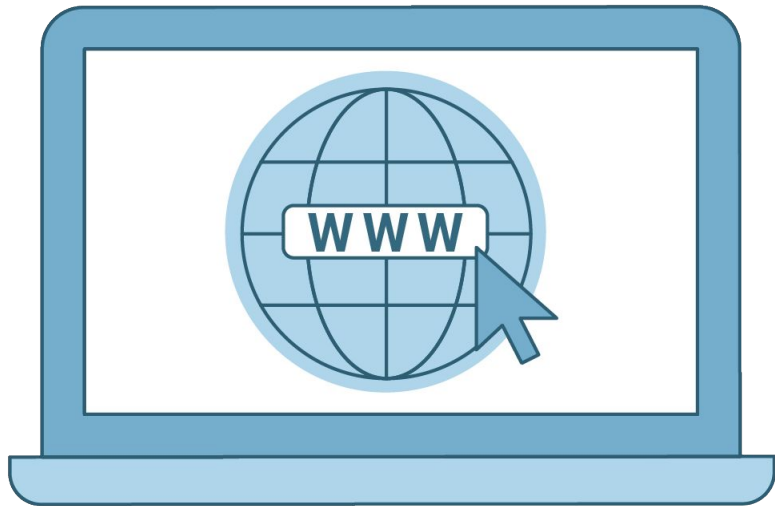


URLs

Uniform Resource Locators

- A URL is the full address of a resource on the internet.
- Like file structures, URLs have a specific syntax indicating where to find the exact resource.

`scheme://subdomain.domain.TLD/path/filename`





URL Syntax

Uniform Resource Locators syntax

https

(Hypertext Transfer Protocol) is a scheme indicating a file on the internet.

facebook

is the primary domain.

/photos/

is the path where the resource is located.

https

://

www

.

facebook

.com

/photos/

catpicture.jpg

www

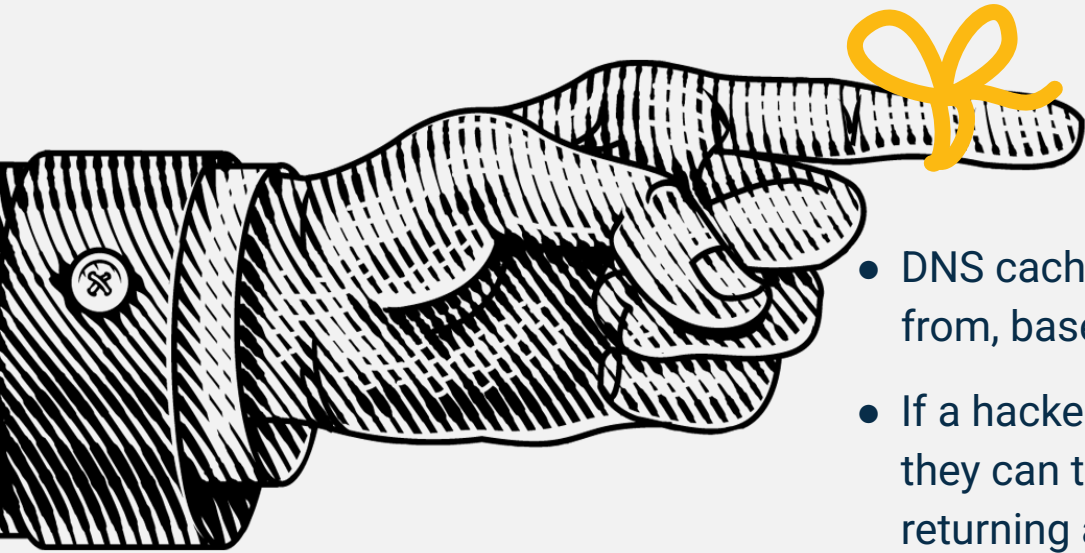
is a subdomain of facebook.com.

.com

is the TLD, or top-level domain.

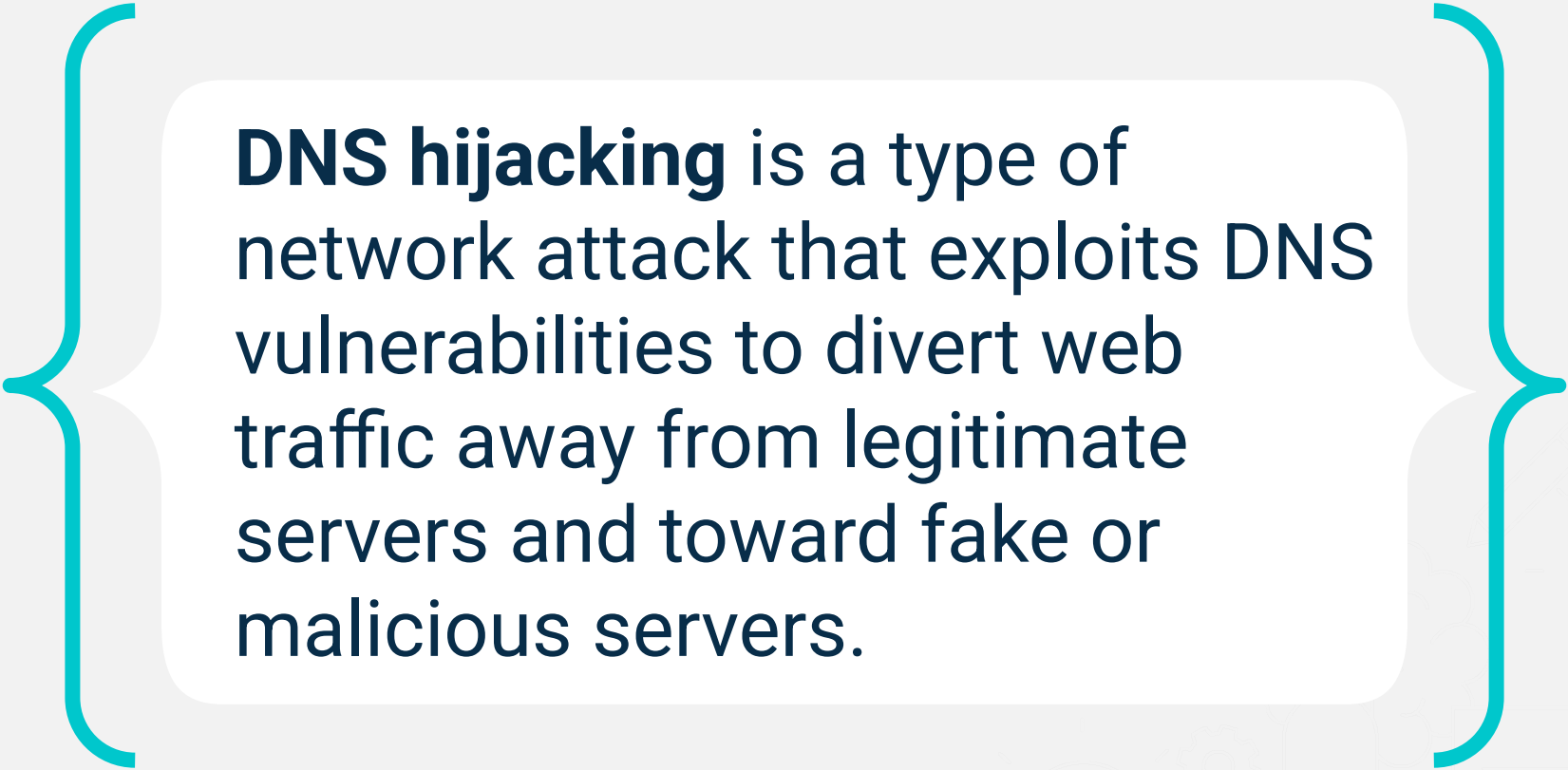
catpicture.jpg

is the resource or file being requested.



Remember

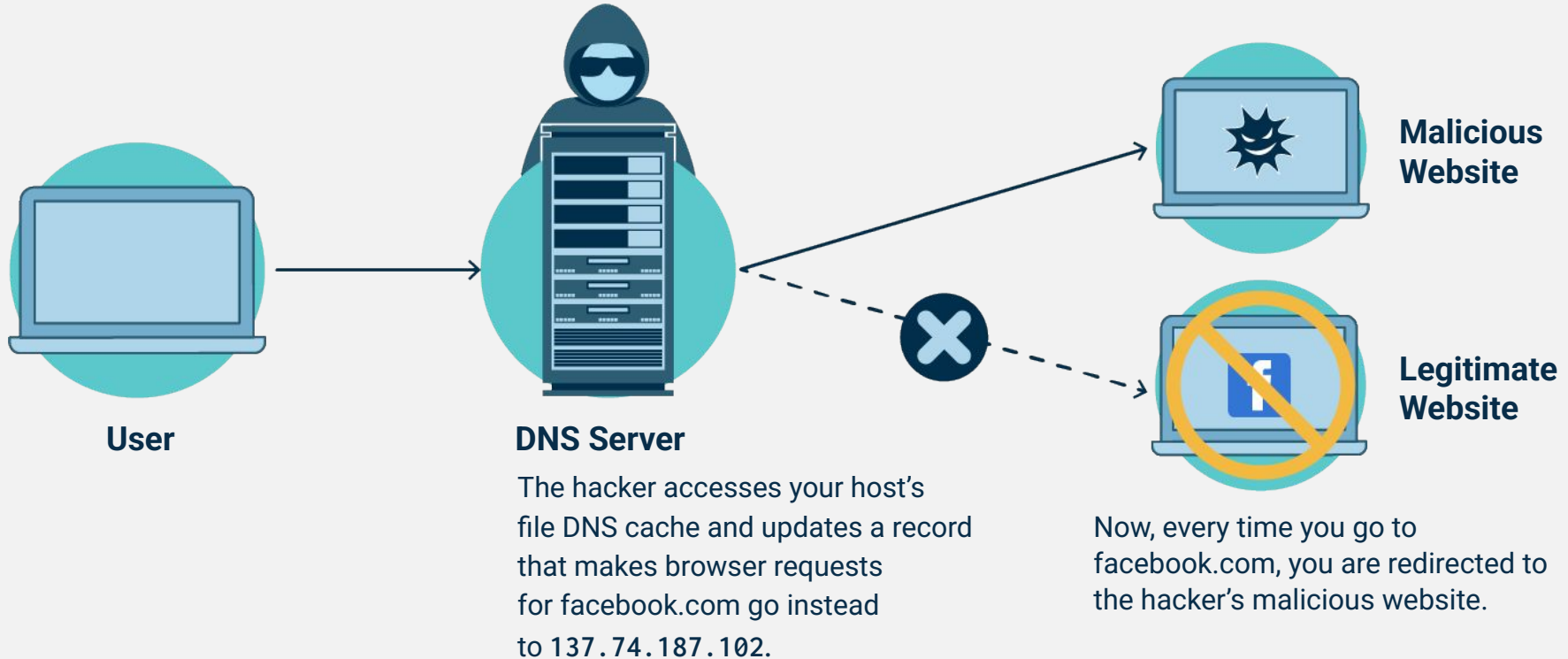
- DNS caches indicate where to request the resources from, based on the domain being accessed.
- If a hacker is able to manipulate the DNS cache, they can trick and exploit a user's request by returning a domain or resource that was not originally requested.

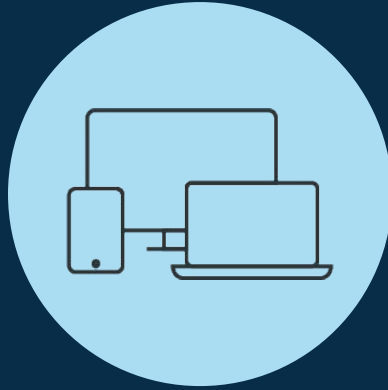


DNS hijacking is a type of network attack that exploits DNS vulnerabilities to divert web traffic away from legitimate servers and toward fake or malicious servers.

DNS, URLs, and Security

Example: A hacker owns a malicious site located at the IP **137.74.187.102**.





Instructor Demonstration

DNS hijacking



Activity

DNS hijacking

In this activity, you will continue to play the role of a security analyst at Acme Corp.

- Your task is to create and test a DNS spoof record that will redirect any hacker trying to visit acmetradesecrets.org to another website.

Suggested Time
20 minutes





Time's Up!
Let's review



Questions?





Next Class

Ports, protocols, and the OSI model



Thank You